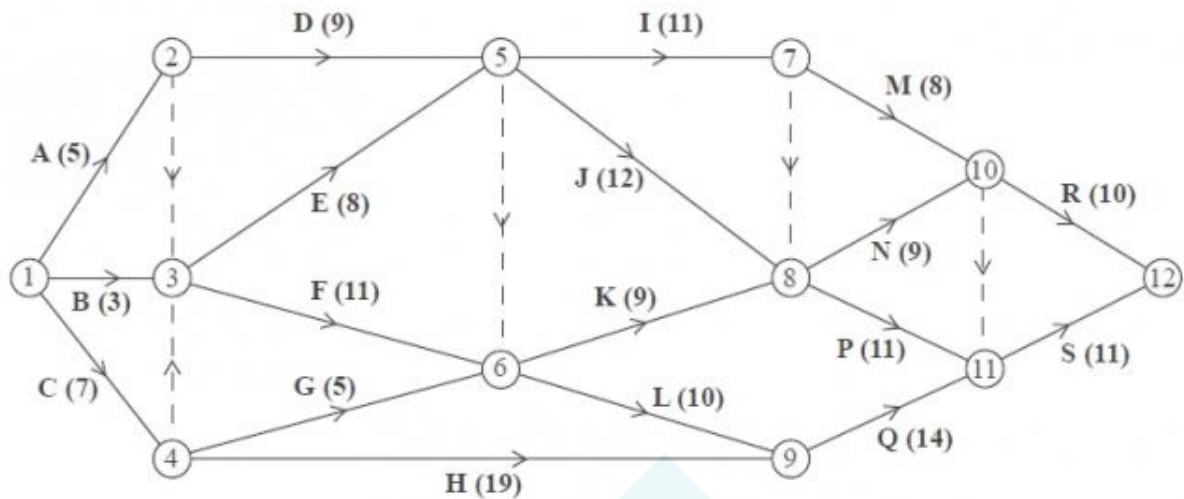


| Given:<br>Due:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Further Maths (Decision)</b><br><b>HW 27</b> | Name:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|---------|----------|----------------------------------|----------|---|----------|---|----------|----------|----------|----------|----------|-------------|----------|-------------|
| <b>Chapter 8: Critical Path Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <b>You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.</b>                                                                                                                                                                                                                                                                                                                                      |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| 91% (A*)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 76% (A)                                         | 66% (B)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 56% (C) | 50% (D) | 40% (E) |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <i>Useful resources:</i> Your notes from recent lessons & your text book                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <b>Questions - do these on lined paper, or use the answer booklet where indicated</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <b>Q1.</b> (a) Draw the activity network described in this precedence table, using activity on arc and exactly two dummies.<br><br>(b) Explain why each of the two dummies is necessary.                                                                                                                                                                                                                                                                                                                                                |                                                 | <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="padding: 5px;">Activity</th> <th style="padding: 5px;">Immediately preceding activities</th> </tr> </thead> <tbody> <tr> <td style="text-align: center; padding: 5px;"><i>A</i></td> <td style="text-align: center; padding: 5px;">-</td> </tr> <tr> <td style="text-align: center; padding: 5px;"><i>B</i></td> <td style="text-align: center; padding: 5px;">-</td> </tr> <tr> <td style="text-align: center; padding: 5px;"><i>C</i></td> <td style="text-align: center; padding: 5px;"><i>A</i></td> </tr> <tr> <td style="text-align: center; padding: 5px;"><i>D</i></td> <td style="text-align: center; padding: 5px;"><i>B</i></td> </tr> <tr> <td style="text-align: center; padding: 5px;"><i>E</i></td> <td style="text-align: center; padding: 5px;"><i>B, C</i></td> </tr> <tr> <td style="text-align: center; padding: 5px;"><i>F</i></td> <td style="text-align: center; padding: 5px;"><i>B, C</i></td> </tr> </tbody> </table> |         |         |         | Activity | Immediately preceding activities | <i>A</i> | - | <i>B</i> | - | <i>C</i> | <i>A</i> | <i>D</i> | <i>B</i> | <i>E</i> | <i>B, C</i> | <i>F</i> | <i>B, C</i> |
| Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Immediately preceding activities                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <i>A</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <i>B</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <i>C</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>A</i>                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <i>D</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>B</i>                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <i>E</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>B, C</i>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <i>F</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>B, C</i>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| <b>Q2.</b> A project is modelled by the activity network shown in Figure 5. The activities are represented by the arcs. The number in brackets on each arc gives the time, in hours, to complete the activity. Some of the early and late times for each event are shown.<br><br>(a) Calculate the missing early and late times and hence complete Diagram 1 in your answer book.<br><br>(b) Calculate the total float on activities D, G and I. You <b>must</b> make your calculations clear.<br><br>(c) List the critical activities. |                                                 | <p style="text-align: center;"><b>Figure 5</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |
| Each activity requires one worker.<br><br>(d) Calculate a lower bound for the number of workers needed to complete the project in the minimum time.                                                                                                                                                                                                                                                                                                                                                                                     |                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |         |         |          |                                  |          |   |          |   |          |          |          |          |          |             |          |             |

**Q3.** A project is modelled by the activity network shown in Figure 5. The activities are represented by the arcs. The number in brackets on each arc gives the time, in days, to complete the activity. Each activity requires one worker. The project is to be completed in the shortest possible time.



**Figure 5**

[The sum of the duration of all activities is 172 days]

- (a) Complete Diagram 1 in the answer book to show the early event times and late event times. (4)
  - (b) Calculate the total float for activity M. You must make the numbers you use in your calculation clear. (2)
  - (c) For each of the situations below, explain the effect that the delay would have on the project completion date. (2)
    - (i) A 2 day delay on the early start of activity P.
    - (ii) A 2 day delay on the early start of activity Q.
  - (d) Calculate a lower bound for the number of workers needed to complete the project in the shortest possible time. (1)
- Diagram 2 in the answer book shows a partly completed cascade chart for this project.
- (e) Complete the cascade chart. (4)
  - (f) Use your cascade chart to determine a second lower bound on the number of workers needed to complete the project in the shortest possible time. You must make specific reference to times and activities. (2)
  - (g) State which of the two lower bounds found in (d) and (f) is better. Give a reason for your answer. (2)

**END OF HOMEWORK**

**Total = 35 marks**

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.

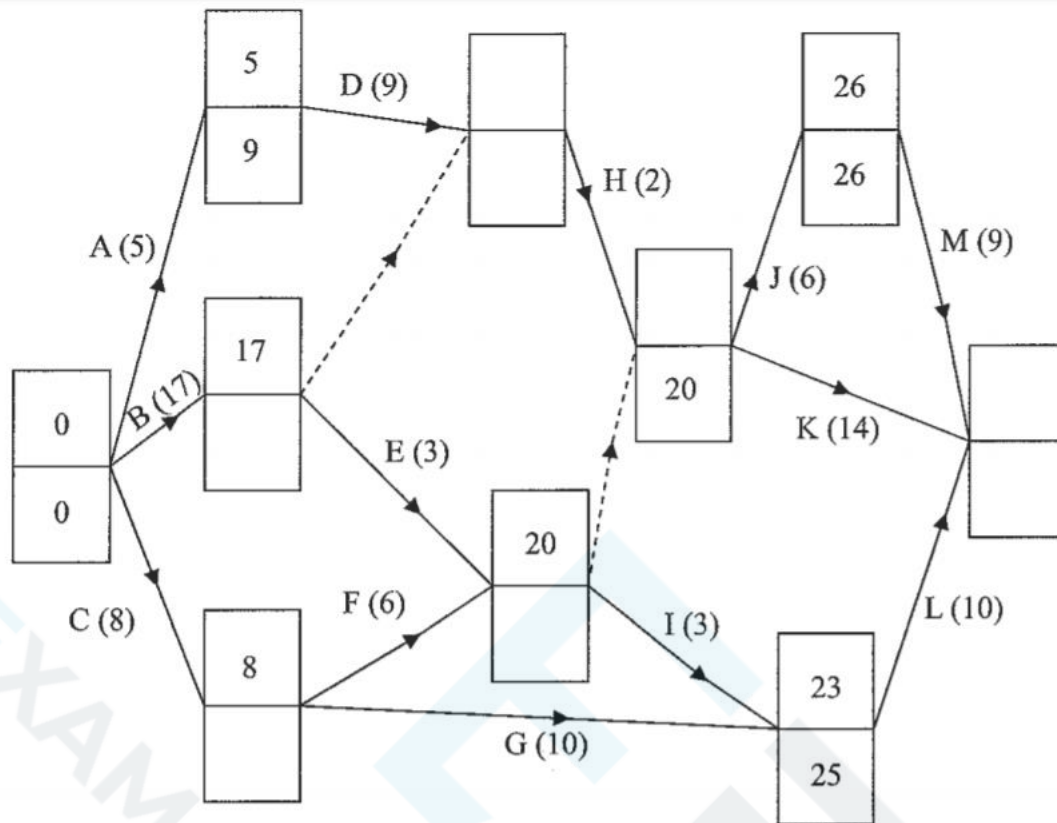
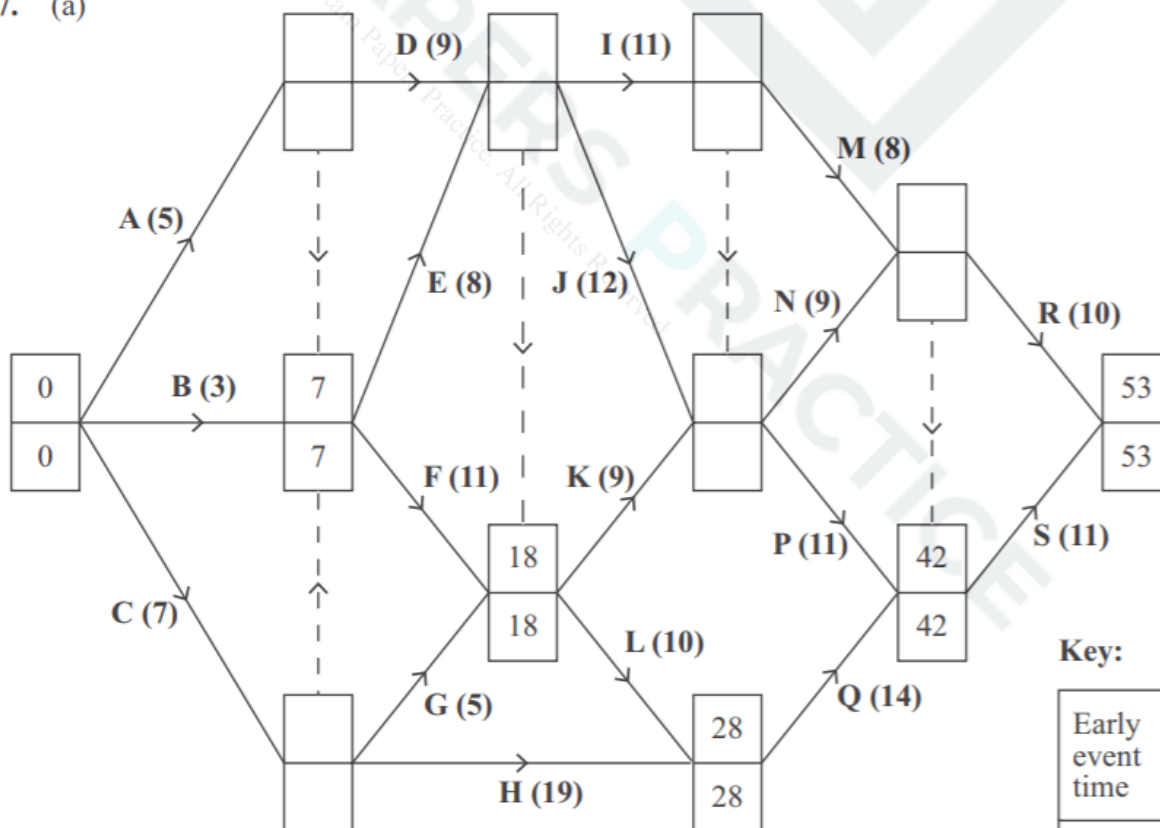


Diagram 1

Q3.

7. (a)



Key:

Early  
event  
time

Late  
event  
time

Diagram 1



### Diagram 2